BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

BIOLOGY

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

BOTANY

SECTION 'A' (Multiple Choice Questions) (8)

> Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed.)

> Note: TIME: 2 Hours — (40 Marks) — BOTANY 2025 (Science Pre-Medical Groups)

Q. SECTION 'A' (Multiple Choice Questions) (8)

> Note: NOTE: i) This section consists of 16 part questions and all are to be answered. Each part question carries ½ mark. ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

Q. 1 Choose the correct answer for each from the given options:

Qi. Process where information present on gene is used to produce a functional product by living organism is called:

(A) Transcription
(B) Translation
(C) Gene expression
(D) Mutation

Qii. It is a condition of true colour blindness:

(A) Monochromacy
(B) Dichromacy
(C) Trichromacy
(D) Haemophilia

Qiii. Organs with similar functions but different in their anatomical features are termed as:

(A) Analogous organs
(B) Homoglous organs
(C) Exterior organs
(D) Interior organs

Qiv. Conversion of nitrates into nitrogen is called:

(A) Nitrification
(B) Denitrification
(C) Ammonification
(D) Nitrogen fixation

Qv. The water cycle begins with the process of:

(A) Transpiration
(B) Combustion
(C) Evaporation
(D) Decomposition

Qvi. The protein that prevents the rebinding of complementary DNA strand during replication fork is called:

(A) DNA Helicase
(B) Ligase
(C) SSB (Single Stranded Binding Protein)
(D) Primase

Qvii. Ecological succession occurring in Arid environments like deserts, mountain and dumes is known as:

(A) Hydrarch
(B) Xerarch
(C) Secondary
(D) Primary

Qviii. This type of culture is used to produce disease free plants from infected plant material:

(A) Meristem culture
(B) Embryo culture
(C) Ovary culture
(D) Anther culture

Qix. This type of cell culture involves the isolation of cells directly from animal tissues and their growth in vitro:

(A) primary cell culture
(B) cell line culture
(C) 3D culture
(D) cloning

Qx. The second stage of Xerarch stage succession is:

(A) Crustose-Lichens stage
(B) Foliase-Lichens stage
(C) Herb stage
(D) Moss stage

Qxi. Mutation in which genes are mutated in normal pattern without any harmful effect:

(A) Positive mutation
(B) Negative mutation
(C) Neutral mutation
(D) Linkage

Qxii. The Endosymbiotic theory was suggested by:

(A) Lynn Margulis
(B) Darwin
(C) Lamarck
(D) Malthus

Qxiii. The gene which interferes of masks the phenotype of the non-locus gene is called:

(A) Mutant gene
(B) Epistatic gene
(C) Pleiotropy
(D) Sex-linked

Qxiv. The alleles of holandric genes is located on:

(A) X-chromosome
(B) Y-chromosome
(C) Both X and Y chromosomes
(D) Autosome

Qxv. According to the International Society of Blood Transfusion, the number of system of blood groups are:

(A) 5
(B) 10
(C) 15
(D) more than 30

Qxvi. Transgenic plant used in biofuel production is:

(A) Cord grass
(B) Lemon grass
(C) Switch grass
(D) Glasswort

SECTION 'B' + Section C (Subjective)

Q. SECTION "B" (Short Answer Question) Marks: 16

> Note: 2. Answer any EIGHT part questions. Each questions carries two marks.

Q. Section C (Detailed -Answer Question) (16)

> Note: NOTE: Answer any TWO questions from this section. All questions carry equal marks.

ZOOLOGY

SECTION 'A' (Multiple Choice Questions) (9)

> Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed.)

> Note: TIME: 2 Hours — (45 Marks) — ZOOLOGY 2025 (Science Pre-Medical Groups)

Q. SECTION 'A' (Multiple Choice Questions) (9)

> Note: NOTE: i) This section consists of 18 part questions and all are to be answered. Each question carries ½ mark. ii) Do not copy the part questions in your answerbook. Write only the answer in full against the proper number of the question and its part.

Q. 1 Choose the correct answer for each from the given options:

Qi. This belongs to endothermic:

(A) Frog
(B) Human
(C) Lizard
(D) Fish

Qii. Kidney secrete this hormone for maintaining of blood pressure:

(A) Erythropoietin
(B) Leukopoietin
(C) Renin
(D) Thrombopoietin

Qiii. A bone lies in between skull and post cranial skeleton is called:

(A) Malleus
(B) Cervical bone
(C) Hyoid bone
(D) Cranial bone

Qiv. In vitro fertilization (IVF) is a method used for:

(A) Treat infertility
(B) Enhance hormones
(C) Prevents STDs
(D) Enhance Enzymes

Qv. It is controlled by medulla oblongata in the brain:

(A) Vision
(B) Memory
(C) Balance
(D) Breathing

Qvi. Goiter is caused by the deficiency of:

(A) Calcium
(B) Iodine
(C) Potassium
(D) Sodium

Qvii. Glycogen is converted into glucose by the action of this hormone:

(A) Insulin
(B) Glucagon
(C) Calcitonim
(D) Cortisol

Qviii. This is related to abnormal or degenerative changes in vertebrae:

(A) Sciatica
(B) Spondylosis
(C) Disc slip
(D) Simple fracture

Qix. Part of human brain Hippocampus name from its shape because it resembles with:

(A) Sea lion
(B) Sea shell
(C) Sea horse
(D) Horse shoe

Qx. Syphilis is a sexually transmitted disease caused by:

(A) Virus
(B) Bacteria
(C) Protozoan
(D) Yeast

Qxi. The number of muscles in human body is:

(A) 300
(B) 400
(C) 500
(D) 600

Qxii. The end of fertility in female is called:

(A) Menarch
(B) Menses
(C) Menopause
(D) Ovulation

Qxiii. It trigger the release of Testosteron into the blood:

(A) PSH
(B) LH
(C) PH
(D) Inhibin

Qxiv. Degeneration of dopamine in neuron caused by this:

(A) Alzheimer's disease
(B) Parkinson's disease
(C) Huntington's disease
(D) Epilepsy

Qxv. The hormone that prevents urination and dehydration is:

(A) Oxytocin
(B) T3
(C) ADH
(D) ACTH

Qxvi. Over secretion of cortisol causes:

(A) Addison's disease
(B) Crushing's disease
(C) Acromegaly
(D) Diabetes

Qxvii. If the body is injured and bleeds, this begin:

(A) Positive feedback
(B) Negative feedback
(C) Both positive and negative feedback
(D) End of feedback

Qxviii. This is caused due to the deficiency of insulin:

(A) High blood pressure
(B) Diarrhea
(C) Diabetes
(D) Anaemia

SECTION 'B' + Section C (Subjective)

Q. SECTION "B" (Short Answer Question) Marks: 18

> Note: NOTE: Attempt any NINE part questions in all. Select FIVE Reasoning Questions and FOUR Non-Reasoning Questions. All part questions carry equal marks. Note: Answer any Nine part questions in all. Select five Reasoning Questions and four Non-Reasoning Questions. All part questions carry equal marks.

Q. Section C (Detailed -Answer Question) (18)

> Note: NOTE: Attempt any TWO question from this section. All questions carry equal marks.